Zain Khan

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Education

University of Waterloo — Waterloo, ON

Bachelor of Applied Science, Computer Engineering — *Sep. 2025 – Apr. 2030*

- Awarded over **$40,000** in merit-based scholarships, including the **University of Waterloo Presidential Scholarship**.

Technical Skills

Languages: TypeScript, JavaScript, Python, C/C++, C#, Java, Ruby, SQL, HTML/CSS
Frameworks: React, Redux, Next.js, Node.js, Express, Django, FastAPI, .NET/WPF, ROS 2, TensorFlow, OpenCV
Infrastructure: PostgreSQL, MySQL, MongoDB, Supabase, AWS, Docker, Linux, Git, Vercel, Playwright

Experience

Leap Tools — Toronto, ON

Software Development Intern, Bespoke Team — *May 2026 – Aug. 2026*

- Built Surface Designer end-to-end (TypeScript, React/Redux, Next.js) for the Roomvo visualizer (**80+ countries**), replacing a server-side C++ tile renderer with an in-browser parametric engine (one generic lattice-expansion algorithm) for live editing with no server round-trip; shipped **19** production patterns now live with customers.
- Shipped commerce features across **6** more production apps on Django microservices (HTTP, gRPC, PostgreSQL, AWS): CORS-enabled REST APIs for vendor storefronts, **10**-event PostHog analytics, and a Postgres/Kysely layer.
- Moved the team onto AI coding-agent harnesses with pre-commit gates, ticket to completion; authored **402** tests, including the app's first Playwright E2E suite (**21** specs), and fixed the SSR-hydration flakes breaking it.

Kasperia — Remote

Software Engineer (Contract) — *Jun. 2023 – Apr. 2026*

- Built **KPM (Kasperia Project Manager)**, a C#/.NET WPF desktop app that became the company's internal procurement operations system, running projects end-to-end from bid to supplier quotes, pricing, and order, with AI agents that draft supplier follow-ups and payment emails from live project state.
- Designed KPM's current-state + history data model, separating live fields (price, lead time, PO, payment) from per-aspect histories (quotes, payments, deliveries) so every value traces to its source.
- Rebuilt the client-facing website with reusable React components, improving page load by **30%** and cutting maintenance overhead by **25%**; built SQL/REST data pipelines that cut manual reporting time by **100%**.

WATonomous — Waterloo, ON

Robotics Software Engineer, EVE Autonomy — *Jan. 2026 – Apr. 2026*

- Developed concurrent autonomous robotics software in C++ on ROS 2 Humble, coordinating **5+ real-time nodes** across perception, planning, and control subsystems.
- Implemented costmaps, A* path planning, and Pure Pursuit control, achieving stable point-to-point navigation in cluttered indoor environments with dynamic obstacle avoidance.

Projects

thinkaloud | *Python, faster-whisper, Windows UI Automation, Next.js, TypeScript, Electron, Docker*

- Built a desktop tool that turns an expert narrating a real computer task into **reviewed, validated training data for computer-use agents**: every click, keystroke, and scroll is paired with before/after screenshots, the UI element it hit, and the spoken reasoning behind it.
- Engineered the local data pipeline: word-timed Whisper transcription aligned per sentence to the action it explains, QC for missing reasoning and leaked secrets, and checksummed exports in Claude's computer-use format; a run counts as a positive demonstration only after a reviewer checks it against a pre-set "done when" checklist.

otto | *Next.js, TypeScript, Python, LiveKit WebRTC, Deepgram, Gemini 2.5 Flash, Supabase*

- Shipped an AI voice agent that turns natural speech into automated actions across GitHub, Gmail, and Google Calendar; awarded **1st Place — LiveKit Track** at NexHacks.
- Engineered a LiveKit WebRTC pipeline with Deepgram VAD turn detection and Gemini 2.5 Flash Realtime native audio inference, achieving **sub-second** voice-to-action latency; cut per-request tokens **4000 → 1800 (55%)**.

toothfairy | *Next.js, TypeScript, FastAPI, Python, TensorFlow, Gemini 2.5 Flash, MedSAM2, MCP, Moorcheh AI*

- Built a dental X-ray segmentation pipeline: a U-Net (TensorFlow/Keras) with OpenCV contour analysis isolates **28+ individual teeth** from panoramic radiographs, with MedSAM2 served on Modal GPUs.
- Built **3 specialized agents** on Gemini 2.5 Flash (imaging, clinical notes, treatment) with Pydantic-validated structured outputs across **11 dental conditions**; the treatment agent calls MCP servers for live drug lookups, and per-patient RAG memory via Moorcheh AI carries history across visits.